

Mathematical Foundations of Political Methodology

Robert Gulotty

Associate Professor
Department of Political Science
University of Chicago

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Why Do We Need to Learn Calculus

Broadly to say, without calculus, the First Industrial Revolution will never happen.

And, for political science, calculus is also very important.

To justify this claim, let's see following two examples.

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Example 1: Derivative in Positive Political Economy

Derivative is very important in positive political economy's research.

There are overlaps between positive political economy, American politics, comparative politics and international relation.

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Example 1: Derivative in Positive Political Economy

We consider an example on redistributive distortionary taxation. The example comes from a textbook, Political Economics: Explaining Economic Policy, by Persson and Tabellini.

The individual has quasi-linear preferences:

$$w = c + V(x)$$

where c and x denote individual consumption and leisure, and $V()$ is a well-behaved concave utility function.

The private budget constraint is:

$$c \leq (1 - \tau)l + f$$

where τ denotes the income tax rate, l the individual labor supply, and f a lump sum transfer.

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In addition to that, individuals are subject to the "time constraint":

$$\alpha \geq x + l$$

where α is individual productivity.

We can maximize individual's utility subject to the budget and time constraints.

The first-order condition implies:

$$1 - \tau = \frac{\partial V}{\partial x}(\alpha - l)$$

where a subscript denotes a derivative.

Then, we can get individual's labor supply: $l = \alpha - \frac{\partial V}{\partial x}^{-1}(1 - \tau)$.

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Example 2: Integral in Statistics

Now, suppose we have the SAT score in 2018, and suppose the distribution of SAT score is normal distribution with mean 1000, standard deviation 300.

Now, let's see a picture of this normal distribution.

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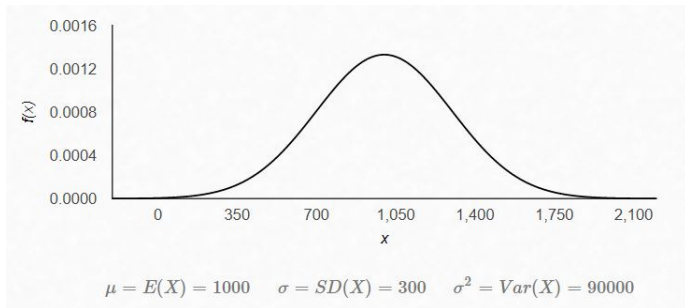
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Suppose that you get a very decent score, say, 1500.

A natural question to ask is: what's the probability of getting a score higher than 1500 in SAT test in 2018.

We need to compute this probability by:

$$Pr(x \geq 1500) = \int_{1500}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 0.04779$$

where $\sigma = 300$, $\mu = 1000$.

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